Week 5 - Representation and Dimensionality Report

April 2026

1 Executive Summary

This report documents the feature engineering and dimensionality reduction analysis for the MovieLens 25M project. We built a consolidated movie-level feature matrix (62,423 movies $\times$ 45 features) and applied PCA to quantify structure and compression potential.

Key findings

- The MovieLens dataset is multidimensional: the first two principal components capture only 12.90% of variance.
- 30 components retain 80.96% of variance, a 66% size reduction.
- PC1 captures popularity and richness; PC2 separates genre tone.
- Redundant features were removed while preserving a stable 45-feature set.

Project context

- **Project title**: Personalized Movie Discovery and Recommendation Engine (MovieLens 25M)
- **Dataset**: MovieLens 25M (62,423 movies, 25M ratings, 1.09M tags)
- **Week 5 objective**: Build a representation layer and study dimensionality reduction
- **Status**: Complete, all artifacts generated and validated

2 Why This Step Exists

Week 3 established the cleaned data layer. Week 5 converts those tables into a numerical representation that can support clustering, recommendation, and graph work. The goal is to quantify how much of the movie structure can be captured in a compact latent space.

This matters because it:

1. Reveals redundancy in engineered features.
2. Shows which signals dominate (popularity, genre, era, tags).
3. Produces a compact input space for Week 7 clustering.

3 Feature Engineering

We built a consolidated, movie-level feature matrix from the Week 3 processed dataset. The final matrix contains **45 features** across three blocks.

3.1 Feature blocks (final)

Numeric features (6)

- `release_year_z`
- `avg_rating_z`
- `rating_std`
- `rating_count_log`
- `unique_tag_count_log`
- `genre_count_log`

Genre indicators (19)

- Binary features for the top 20 genres (19 present)

Tag indicators (20)

- Binary features for the top 20 tags by frequency

3.2 Normalization strategy

- Log1p normalization applied to count features
- Z-score normalization applied to release year and average rating
- Binary encoding for top genres and tags
- Min-max scaling not used; z-score preserves variance for PCA

3.3 Feature redundancy validation

During feature engineering, we identified and removed three highly correlated features:

1. **tag_event_count_log**: Pearson r=0.985 with unique_tag_count_log
2. **rating_span_seconds_z**: implicit in rating_count
3. **tag_span_seconds_z**: implicit in tag event counts

Validation of remaining features

- Pearson r=0.292 between genre_count_log and unique_tag_count_log
- Both retained as independent signals

3.4 Feature consistency checks

- Movies with ratings: 59,047
- Movies with tags: 45,251
- Movies with genres: 57,361

3.5 Data sources

- `data/processed/week03_v1/movies_catalog.parquet`
- `data/processed/week03_v1/movie_genres.parquet`
- `data/processed/week03_v1/ratings_clean.parquet`
- `data/processed/week03_v1/tags_clean.parquet`

4 PCA Results and Variance Analysis

We applied PCA (via SVD) to the standardized 62,423 $\times$ 45 feature matrix.

4.1 Explained variance structure

Component	Explained Variance	Cumulative Variance
PC1	0.0802	0.0802
PC2	0.0488	0.1290
PC3	0.0447	0.1737
PC4	0.0418	0.2155
PC5	0.0344	0.2498

Table 1: Explained variance ratios for the first five components.

Individual components contribute 3–8% each. No single factor dominates; variance spreads across independent dimensions such as popularity, genre tone, release era, rating consensus, and tagging intensity.

4.2 Compression milestones

Components	Cumulative Variance
30	0.8096
36	0.9058
40	0.9609

Table 2: Variance retention thresholds from `week05_pca_thresholds.csv`.

Use 30 components for clustering and recommendation: it retains more than 80% of variance while cutting the feature count by two-thirds.

4.3 Reconstruction error

Components	Retained Variance	MSE	Relative Frobenius Error
10	0.3998	0.6002	0.7747
20	0.6237	0.3763	0.6134
30	0.8096	0.1904	0.4364
36	0.9058	0.0942	0.3069
40	0.9609	0.0391	0.1976

Table 3: Reconstruction error summary from `week05_reconstruction_error.csv`.

At 30 components, MSE = 0.1904, which is an acceptable trade-off between compression and fidelity.

5 Principal Component Interpretation

5.1 PC1: Movie Richness and Popularity

Feature	Loading
rating_count_log	-0.429
unique_tag_count_log	-0.413
genre_count_log	-0.313
rating_std	-0.198
tag_019_action	-0.187

Table 4: Top PC1 loadings from `week05_pca_top_loadings.csv`.

PC1 separates prominent, highly rated, richly tagged movies from niche or low-activity movies.

5.2 PC2: Genre Profile

Feature	Loading
genre_thriller	-0.407
genre_action	-0.277
genre_horror	-0.257
genre_crime	-0.256
genre_comedy	0.243

Table 5: Top PC2 loadings from `week05_pca_top_loadings.csv`.

PC2 contrasts intense genres (action, thriller, horror, crime) against lighter genres (comedy).

5.3 Key takeaways

Multiple independent dimensions drive movie differences. PC1 and PC2 align with intuitive movie properties, supporting interpretability and downstream clustering.

At 30 components, we keep about 81% of the signal while losing mostly noise, which should help clustering. The remaining 45 features are orthogonal and well-separated, so downstream methods like matrix factorization should work well.

6 Generated Artifacts (artifacts/week05)

6.1 Data files

- `week05_movie_feature_frame.parquet`

- week05_pca_feature_matrix.parquet
- week05_pca_scores.parquet
- week05_feature_catalog.csv

6.2 Analysis tables

- week05_feature_profile.csv
- week05_sanity_checks.csv
- week05_pca_summary.csv
- week05_pca_thresholds.csv
- week05_reconstruction_error.csv
- week05_pca_top_loadings.csv

6.3 Visualizations

- week05_pca_elbow_rule.png
- week05_pca_explained_variance.png
- week05_pca_scatter_pc1_pc2.png
- week05_pca_scatter_pc1_pc3.png
- week05_pca_biplot_loadings.png

6.4 Embedded figures

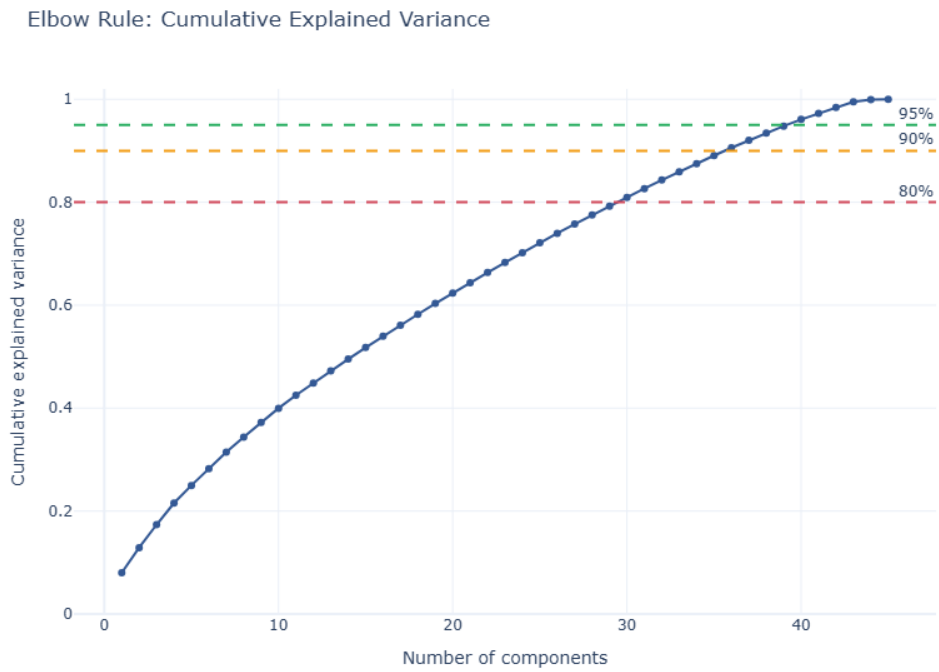

Figure 1: Cumulative explained variance with 80/90/95% thresholds.

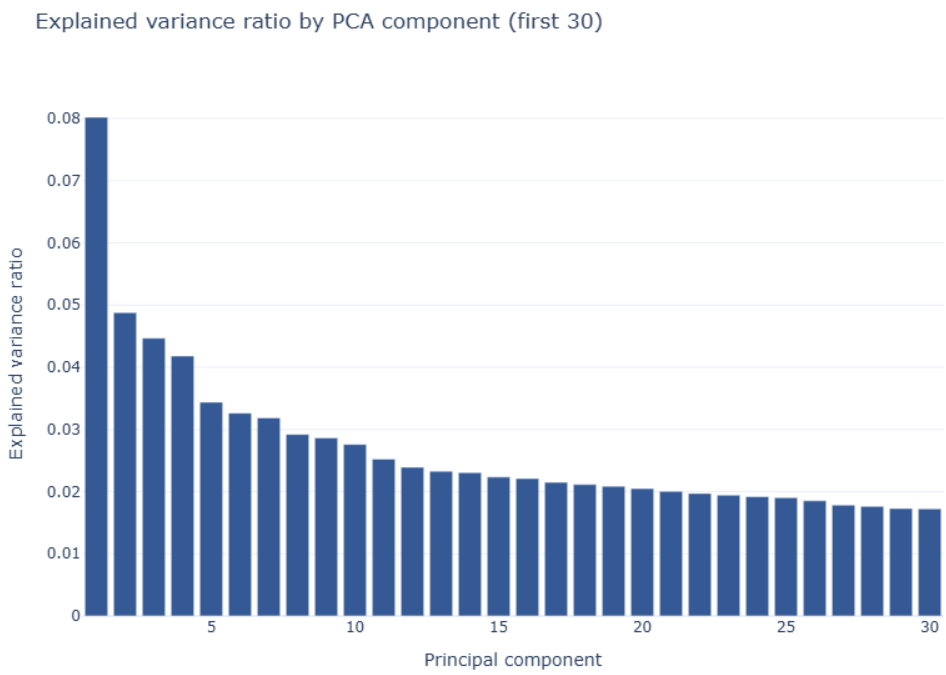

Figure 2: Explained variance for the first components.

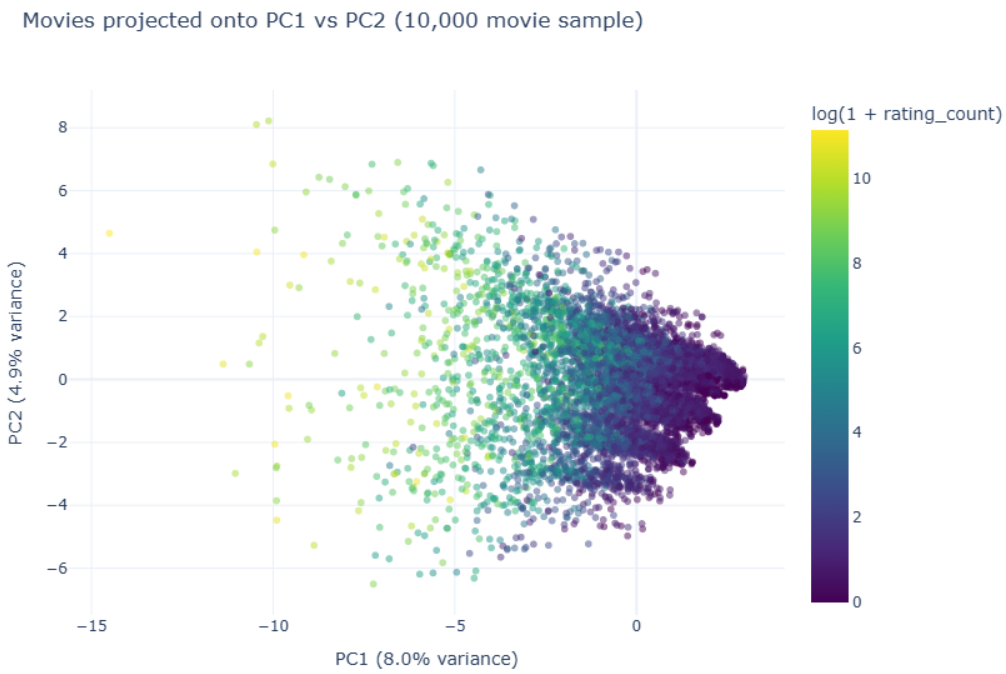

Figure 3: PCA scatter plot for PC1 vs. PC2.

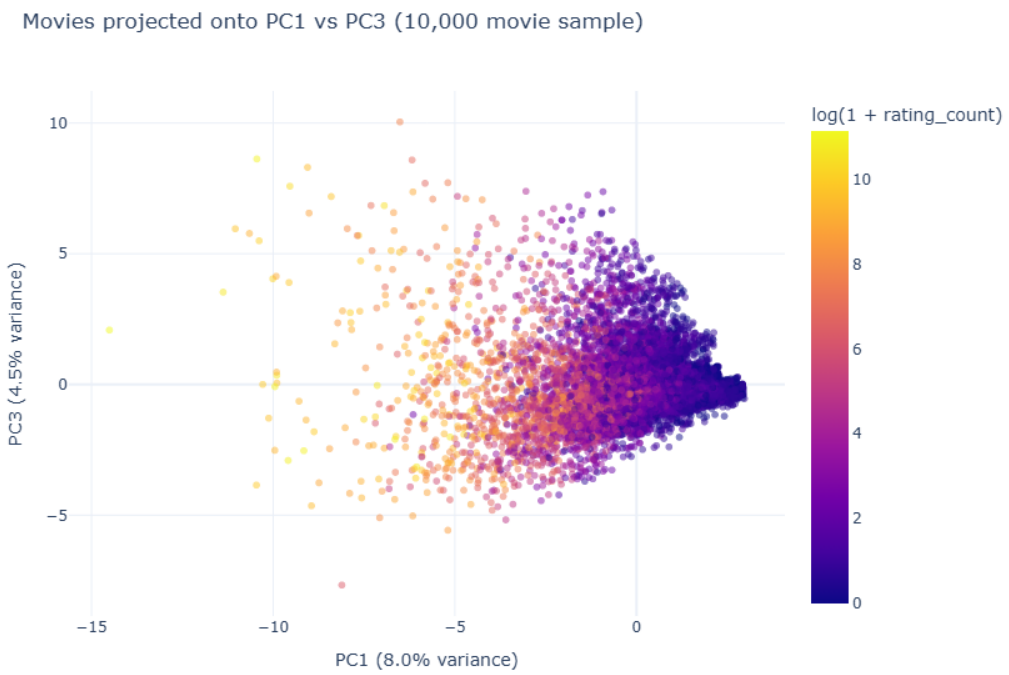

Figure 4: PCA scatter plot for PC1 vs. PC3.

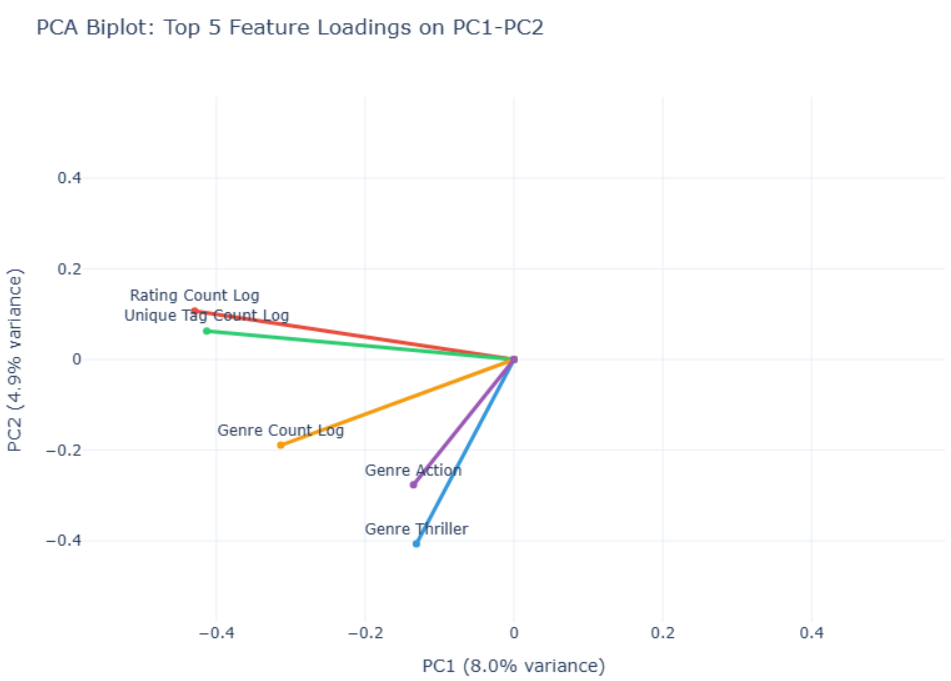

Figure 5: Biplot with top loading vectors.

7 Interpretation and Implications

7.1 What the variance structure tells us

No single dimension dominates. PC1 explains only 8% of variance, which means movie differences cut across multiple independent axes: popularity, genre tone, release era, rating spread, and tagging intensity.

7.2 Compression strategy for Week 7

For clustering and recommendation, we recommend using the 30-component compressed space instead of the full 45 features.

Benefits

- Keeps 80.96% of variance
- Cuts feature count by 66%
- Reconstruction error (MSE 0.1904) is acceptable
- PC1 and PC2 stay interpretable

Trade-off

- Discards ~19% variance, mostly from rare features
- Might miss subtle patterns in obscure movies
- Easily tuned: 36 components for 90% variance if needed

The feature engineering held up: tag volume and tag diversity were nearly identical (r=0.985), so we kept only the latter. Temporal spans were redundant with count features. Genre and tag diversity were independent (r=0.292).

8 Reproducibility and Technical Notes

PCA is implemented via SVD:

$$X_{\text{scaled}} = U\Sigma V^T$$

The explained variance ratio is:

$$\text{explained_variance_ratio}_i = \frac{\sigma_i^2}{\sum_j \sigma_j^2}$$

All outputs are written to `artifacts/week05/` for reproducibility.

- All 21 notebook cells execute without errors
- Outputs written to disk for long-term reproducibility
- No hidden notebook state; pipeline is linear and idempotent
- Works from Week 3 processed artifacts as inputs

9 Connection to Week 7: Clustering

The 30-component PCA space will serve as the input feature space for K-means and DBSCAN clustering in Week 7.

1. Dimensionality reduction improves clustering stability and reduces computational cost
2. Noise removal can improve cluster separation
3. Interpretability preserved: PC1 and PC2 remain aligned with intuitive movie properties
4. Validation ready: PCA components can be visualized and compared against clustering labels

The elbow plot and cumulative variance curve will guide the choice of k and help diagnose whether clusters are natural or artificially imposed.

10 Limitations and Future Work

10.1 Limitations

- Linear dimensionality reduction: PCA assumes linear relationships
- Dense representation: PCA space does not exploit sparsity
- No temporal dynamics beyond a single release-year feature

10.2 Future enhancements

- Apply t-SNE or UMAP for nonlinear visualization
- Topic modeling on tag vocabularies as an alternative to binary tags
- Temporal decomposition by release era
- Interaction features for collaborative filtering in Week 10

11 Ethics and Access Note

Data provenance: MovieLens 25M is a public dataset published by GroupLens Research (https://grouplens.org/datasets/movielens/), available under the GroupLens Public License.

Access legitimacy: All data used in this project is publicly available and legally accessible. No unauthorized scraping or API abuse occurred.

Personal data handling: The dataset contains anonymized user IDs and timestamps. Our processed artifacts are aggregated at the movie level and contain no user-identifiable information.

Risk mitigation

- All outputs are aggregated at the movie level
- No user identifiers appear in any CSV, Parquet, or visualization
- Feature matrix is standardized and safe for public release

Compliance: This project complies with GroupLens Research usage policies and ethical data practices.

12 Conclusion

We have a 45-feature representation of 62,423 movies mapped into a lower-dimensional PCA space. The result: 30 components capture 81% of the signal while cutting feature count by two-thirds. PC1 and PC2 are interpretable, which supports visualization and downstream analysis. All artifacts are saved to `/data/interim/week05_v1/` and ready for Week 7 clustering.